

Data and regression

Last time: class activity

$$\text{performance} = \beta_0 + \beta_1 \text{size} + \beta_2 \text{females} + \beta_3 \text{diversity} + \beta_4 \text{testosterone} \\ + \beta_5 \text{diversity} \cdot \text{testosterone} + \varepsilon$$

$$p = 6$$

The researchers hypothesize that increased group diversity is positively associated with performance when group testosterone levels are low, but increased diversity is negatively associated with performance when testosterone levels are high.

What parameter are we interested in for statistical inference?

$$H_0: \beta_5 = 0$$

$$H_A: \beta_5 < 0$$

$$t\text{-test: } t = \frac{\hat{\beta}_5 - 0}{SE \hat{\beta}_5}$$

$$t_{n-p} \\ \Rightarrow t_{68}$$

95% CI :

$$\hat{\beta}_5 \pm t^* SE \hat{\beta}_5$$

$$n = \# \text{ obs } (74) \\ p = \# \text{ params } (6)$$

Last time: class activity

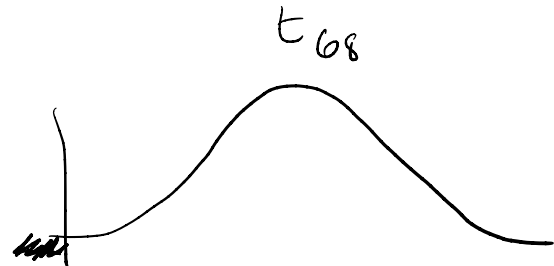
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-8.99931	1.94474	-4.628	1.71e-05	***
size	0.27770	0.16009	1.735	0.0873	.
females	-0.34429	0.20501	-1.679	0.0977	.
diversity	16.12690	3.72741	4.327	5.08e-05	***
testosterone	0.07564	0.01633	4.633	1.68e-05	***
diversity:testosterone	-0.14796	0.03221	-4.594	1.94e-05	***

$\hat{\beta}_5$
 $SE_{\hat{\beta}_5}$

p-value for
 $H_0: \beta_5 = 0$
 $H_A: \beta_5 \neq 0$

$$t = -4.594$$

$$H_0: \beta_5 = 0 \quad H_A: \beta_5 < 0$$



$$-4.594$$

$$p\text{-value} \approx 9.7 \times 10^{-6}$$

95% CI:

$$t^* = 1.995$$

$$-0.148 \pm 1.995 (0.0322)$$

$$= (-0.212, -0.084)$$

Activity: thinking about data

Work on the activity (handout).

Activity

Consider dividing a group into two subgroups. *One possible* (like SSE)
measure of group homogeneity is

$$\text{Homogeneity} = \frac{\sum_{j=1}^p \sum_{k=1}^2 m_k (\bar{x}_{.jk} - \bar{x}_{.j.})^2}{\sum_{j=1}^p \sum_{k=1}^2 \sum_{i=1}^{m_k} (x_{ijk} - \bar{x}_{.j.})^2}$$

← variability after dividing into subgroups

← overall variability (like

p = # demographic variables (Age, Gender, etc.) SST_{Total}

k = Subgroup (subgroup 1 vs. subgroup 2)

m_k = # individuals in subgroup k ($m_1 + m_2 = n$)

x_{ijk} = value of variable j for person i in subgroup k

$\bar{x}_{.j.}$ = mean value of variable j

$\bar{x}_{.jk}$ = mean variable j in subgroup k 5/7

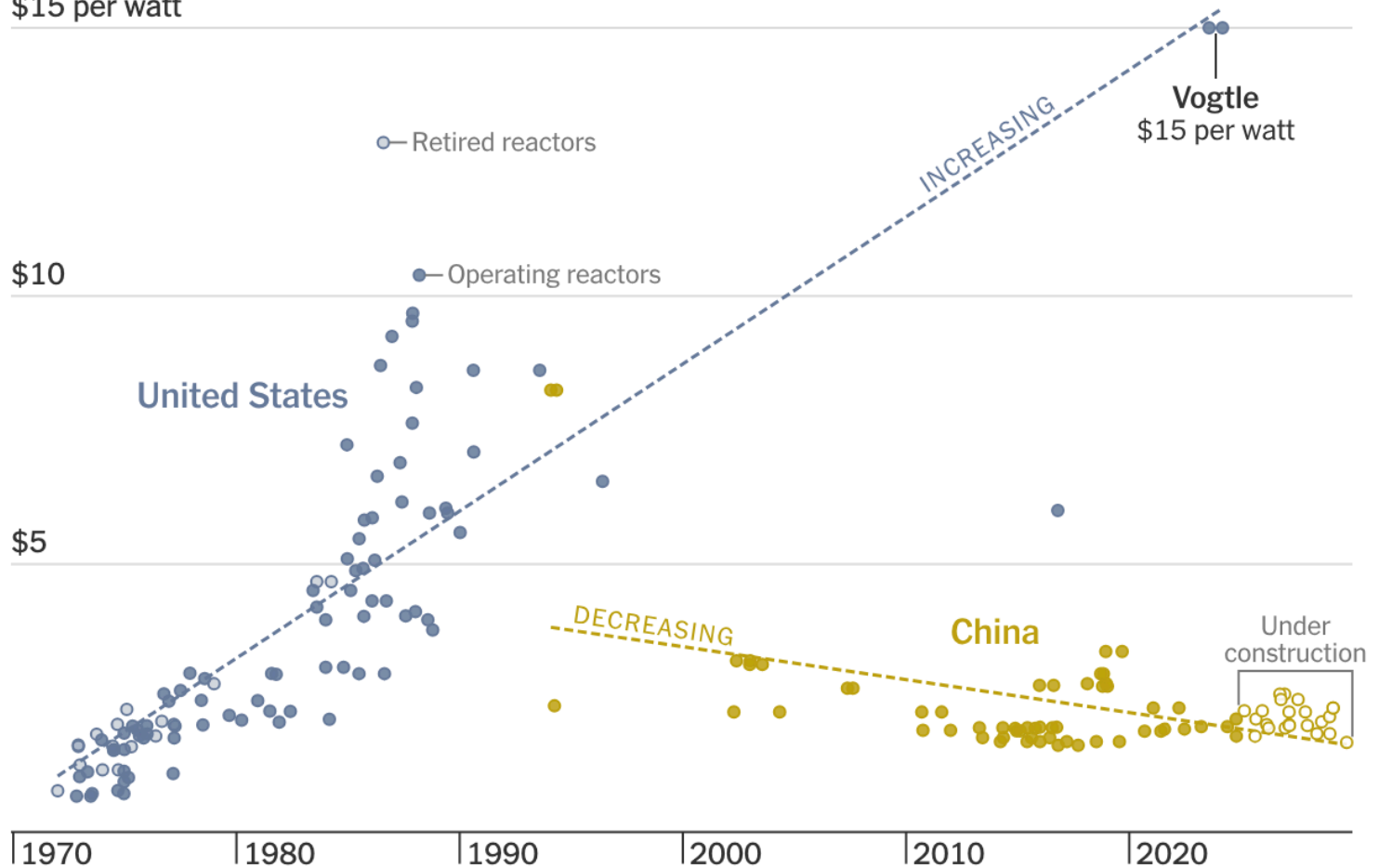
Activity: thinking about data

Work on the class activity (handout).

Activity

Construction costs of nuclear reactors

\$15 per watt



Hover to explore the data